

❖ **Brownian Motion 2 – Geometric BM**

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Geometric Brownian Motion (GBM) is the fundamental stochastic model for in mathematical finance:

- stock prices,
- exchange rates,
- population growth under randomness,
- many physical growth processes.

Geometric Brownian Motion (GBM)

We define **Geometric Brownian Motion** by

$$X(t) = X(0) e^{\mu t + \sigma B(t)}, t \geq 0,$$

where

- $B(t)$ is a standard Brownian motion,
- $X(0) > 0$ is the initial value,
- $\mu \in \mathbb{R}$ is the *drift rate*,
- $\sigma > 0$ is the *volatility* parameter (so σ^2 is the variance rate).

Review: Facts about Brownian Motion for all computations for GBM

Let $B(t)$ be standard BM. Then:

- **Start at zero** $B(0) = 0$.
- **Normal distribution at fixed time** $B(t) \sim \mathcal{N}(0, t)$. Hence

$$B(t) = \sqrt{t} Z, \quad Z \sim \mathcal{N}(0, 1).$$

- **Independent increments**

For $0 \leq s < t$,

$B(t) - B(s)$ is independent of $\{B(u): u \leq s\}$.

- **Stationary increments (memoryless in time)**

$$B(t) - B(s) \sim \mathcal{N}(0, t - s) = \sqrt{t - s} Z.$$

- **Moment generating function of standard normal**

If $Z \sim \mathcal{N}(0, 1)$, then

$$\mathbb{E}[e^{tZ}] = e^{t^2/2}.$$

Distribution of $X(t)$

Using $B(t) = \sqrt{t} Z$, we can write

$$X(t) = X(0) e^{\mu t + \sigma \sqrt{t} Z}.$$

Thus:

$$\ln X(t) = \ln X(0) + \mu t + \sigma \sqrt{t} Z.$$

So:

$$\ln X(t) \sim \text{Normal}(\ln X(0) + \mu t, \sigma^2 t).$$

Hence: $X(t)$ is **lognormally distributed**.

Basic Properties of GBM

(a) Positivity. Since it is an exponential,

$$X(t) > 0 \text{ for all } t \geq 0.$$

(b) Representation at fixed time For all $t > 0$,

$$X(t) = X(0)e^{\mu t + \sigma\sqrt{t}Z}, \quad Z \sim \mathcal{N}(0,1).$$

This shows explicitly how randomness enters through a normal variable.

(c) Independent *fractional* increments

For $0 \leq s < t$,

$$\frac{X(t)}{X(s)} = \exp\left(\mu(t-s) + \sigma(B(t) - B(s))\right).$$

Since $B(t) - B(s)$ is independent of the past,

$$\frac{X(t)}{X(s)} \text{ is independent of } \{X(u): u \leq s\}.$$

This is the multiplicative analogue of independent increments for Brownian motion.

(d) Stationary fractional increments (memoryless)

Because

$$B(t) - B(s) \sim B(t - s),$$

we get

$$\frac{X(t)}{X(s)} \stackrel{d}{=} \exp(\mu(t - s) + \sigma B(t - s)) = \frac{X(t - s)}{X(0)}.$$

So:

$$\frac{X(t)}{X(s)} \stackrel{d}{=} \frac{X(t - s)}{X(0)}$$

This means the *distribution of relative growth* over any interval depends only on its length.

Moments of $X(t)$

Mean: $m(t) = \mathbb{E}[X(t)] = X(0)e^{\mu t} \mathbb{E}[e^{\sigma B(t)}]$

$$= X(0)e^{\mu t} \mathbb{E}\left[e^{\sigma\sqrt{t}Z}\right]$$
$$= X(0)e^{\mu t} e^{\frac{1}{2}\sigma^2 t}$$
$$= X(0) e^{(\mu + \frac{1}{2}\sigma^2)t}$$

Higher Moments: For any integer $n \geq 1$:

$$m_n(t) = E[X(t)^n] = X(0)^n e^{n\mu t} \mathbb{E}[e^{n\sigma B(t)}]$$
$$= X(0)^n e^{n\mu t} e^{\frac{1}{2}n^2\sigma^2 t}$$
$$= X(0)^n e^{n\mu t + \frac{1}{2}n^2\sigma^2 t}$$

- The mean grows faster than $e^{\mu t}$ because of randomness.
- The extra term $\frac{1}{2}\sigma^2$ is the **Ito correction** (appears naturally here via the mgf).
- This shows the moments grow extremely fast when σ is large

Covariance Function

For $s < t$,

$$R(s, t) = E[X(s)X(t)]$$

$$= E \left[X(s)^2 \frac{X(t)}{X(s)} \right]$$

$$= E[X(s)^2] E \left[\frac{X(t)}{X(s)} \right]$$

Use independence of increments

$$= E[X(s)^2] E \left[\frac{X(t-s)}{X(0)} \right]$$

By memoryless

$$= E[X(s)^2] \frac{1}{X(0)} \mathbb{E}[X(t-s)]$$

$$= m_2(s) X(0)^{-1} m(t-s)$$

Here $m_2(s) = X(0)^2 e^{2\mu s + 2\sigma^2 s}$

$$m(t-s) = X(0) e^{(\mu + \frac{1}{2}\sigma^2)(t-s)}$$

Interpretation

GBM is the multiplicative analogue of Brownian motion:

Brownian Motion	Geometric Brownian Motion
Additive noise	Multiplicative noise
$B(t) \sim N(0, t)$	$\ln X(t) \sim N(\ln X(0), \mu t \sigma^2 t)$
Independent increments	Independent fractional increments
Stationary increments	Stationary growth ratios
Gaussian	Lognormal

<https://drive.google.com/file/d/1Vhi4pnD3PZIsIrLVlpq2TfB6wyQu1cuJ/view?usp=sharing>

GBM stock-price model

Stock prices are often modeled by **geometric Brownian motion**

- Historical data for many stocks show long-term exponential growth or decline.
- Stock prices cannot be negative, and geometric Brownian motion always produces positive values.
- Let Y_t denote the price of a stock after t days. Since the price on one day is usually close to the price on the next day (under normal market conditions), stock prices themselves are not independent.
- However, the *percent changes* in price from day to day,

$$\frac{Y_t}{Y_{t-1}}, \quad t = 1, 2, \dots,$$

may reasonably be modeled as IID random variables.

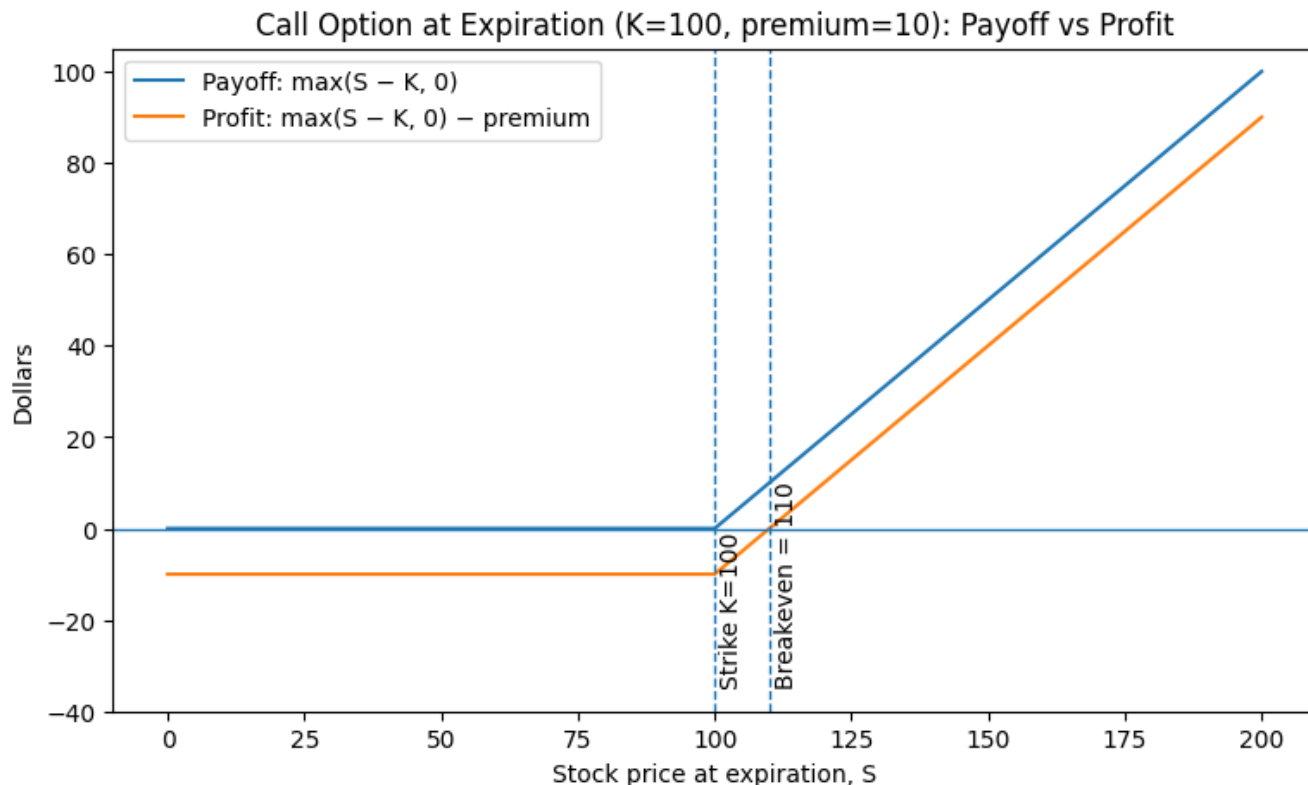
A criticism of the geometric Brownian motion model is that it does not account for extreme events, such as the stock market crash of October 19, 1987, when the world's stock markets lost more than 20% of their value within a few hours.

In stock prices, the standard deviation parameter σ is called the **volatility**.

Example. Now suppose that the stock XYZ currently sells for \$80 per share and follows a geometric Brownian motion with:

- drift parameter $\mu = 0.10$,
- volatility $\sigma = 0.50$.

Find the probability that in 90 days the price of XYZ will rise to at least \$100.



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Model and given data

Assume the stock price $(Y_t)_{t \geq 0}$ follows **geometric Brownian motion**:

$$Y_t = Y_0 e^{\mu t + \sigma B_t},$$

where B_t is standard Brownian motion, μ is drift, and σ is volatility.

From the example, given

- Current price: $Y_0 = 80$
- Drift: $\mu = 0.10$
- Volatility: $\sigma = 0.50$
- Time horizon: 90 days $\approx t = 1/4 = 0.25$ years
- Target: $\mathbb{P}(Y_{0.25} \geq 100)$

Rewrite the event in terms of Brownian motion

Start from

$$Y_{0.25} = 80 e^{\mu(0.25) + \sigma B_{0.25}}.$$

We want

$$\mathbb{P}(Y_{0.25} \geq 100) = \mathbb{P}(80 e^{\mu(0.25) + \sigma B_{0.25}} \geq 100).$$

Solve for $B_{0.25}$:

$$\mathbb{P}(Y_{0.25} \geq 100) = \mathbb{P}\left(B_{0.25} \geq \frac{\ln(1.25) - \mu(0.25)}{\sigma}\right).$$

$$= \mathbb{P}(B_{0.25} \geq 0.3963). \quad B_{0.25} \sim \mathcal{N}(0, 0.25)$$

$$= \mathbb{P}\left(\frac{B_{0.25}}{0.5} \geq \frac{0.3963}{0.5}\right)$$

$$= \mathbb{P}(Z \geq 0.7926)$$

$$= 1 - \Phi(0.7926) \approx 0.214.$$

Example (Financial options).

An *option* is a financial contract that gives the buyer the right, but not the obligation, to buy a share of stock at a fixed price on a specified future date.

Suppose the stock XYZ currently sells for \$80 per share. An option on XYZ is selling today for \$10. Under the terms of the option, in 90 days you may buy one share of XYZ for \$100. This fixed price \$100 is called the **strike price**.

If you buy the option, your payoff in 90 days depends on the stock price at that time:

- If the price of XYZ in 90 days is greater than \$100, you will exercise the option. You buy the stock for \$100 and immediately sell it at the market price.

If the stock price is $G_{90/365}$, your payoff is

$$G_{90/365} - 100.$$

- If the price of XYZ in 90 days is less than \$100, you will not exercise the option. The option expires worthless and your payoff is 0.

Thus, in all cases, the **payoff** in 90 days is

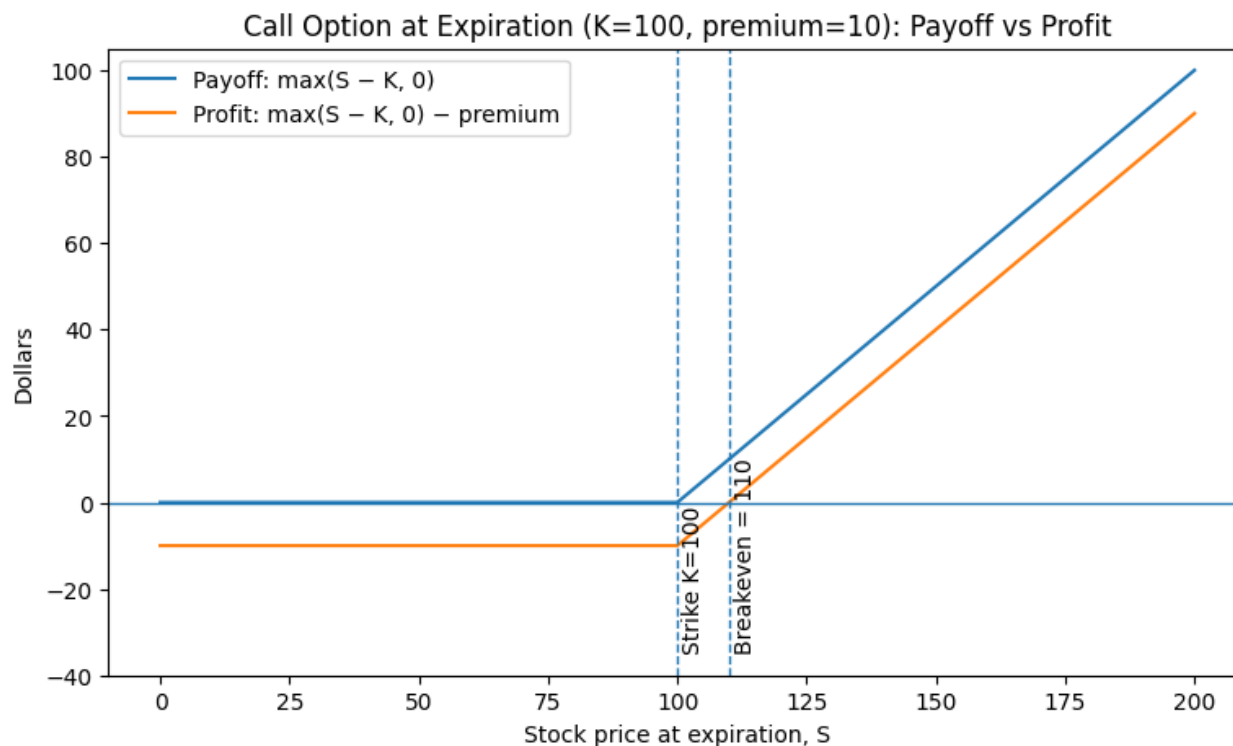
$$\max\{G_{90/365} - 100, 0\}.$$

Your **profit** is this payoff minus the \$10 you paid today for the option.

We are asked to find the **expected payoff**

$$E[\max\{G_{90/365} - 100, 0\}],$$

assuming that the stock price follows a geometric Brownian motion.



Model and notation

- G_0 =current stock price,
- G_t =stock price after t years,
- t = time to expiration (in years),
- K = strike price.

For stock XYZ:

$$G_0 = 80, t = \frac{90}{365}, K = 100.$$

Assume G_t follows a geometric Brownian motion of the form

$$G_t = G_0 e^{\mu t + \sigma B_t},$$

where:

- μ is the drift parameter,
- σ is the volatility,
- B_t is standard Brownian motion.

We want: $E(\max\{G_t - K, 0\})$.

$$E(\max\{G_t - K, 0\}) = E(\max\{G_0 e^{\mu t + \sigma B_t} - K, 0\})$$

$$= \int_{-\infty}^{\infty} \max\{G_0 e^{\mu t + \sigma B_t} - K, 0\} f(x) dx.$$

where $f(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}$ is the pdf of $B_t \sim N(0, t)$

The payoff is positive only when

$$G_0 e^{\mu t + \sigma x} - K > 0 \iff x > \beta,$$

where

$$\beta = \frac{\ln(K/G_0) - \mu t}{\sigma}.$$

Therefore,

$$\begin{aligned} E(\max\{G_t - K, 0\}) &= \int_{\beta}^{\infty} \max\{G_0 e^{\mu t + \sigma B_t} - K, 0\} f(x) dx. \\ &= G_0 e^{\mu t} \int_{\beta}^{\infty} e^{\sigma x} f(x) dx - K \int_{\beta}^{\infty} f(x) dx. \\ &= G_0 e^{t(\mu + \sigma^2/2)} \mathbb{P}\left(Z > \frac{\beta - \sigma t}{\sqrt{t}}\right) - K \mathbb{P}\left(Z > \frac{\beta}{\sqrt{t}}\right) \end{aligned}$$

Reason for the last step:

First integral:

$$\begin{aligned}\int_{\beta}^{\infty} e^{\sigma x} f(x) dx &= \int_{\beta}^{\infty} e^{\sigma x} \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)} dx. \\ &= e^{\sigma^2 t/2} \int_{\beta}^{\infty} \frac{1}{\sqrt{2\pi t}} e^{-(x-\sigma t)^2/(2t)} dx. \\ &= e^{\sigma^2 t/2} P\left(Z > \frac{\beta - \sigma t}{\sqrt{t}}\right).\end{aligned}$$

Second integral:

$$\int_{\beta}^{\infty} f(x) dx = P(B_t > \beta) = P\left(Z > \frac{\beta}{\sqrt{t}}\right),$$

Numerical evaluation for stock XYZ

Recall that $G_0 = 80$, $t = \frac{90}{365}$, $K = 100$.

Assume:

$$\mu = 0.10, \sigma^2 = 0.25 \quad (\sigma = 0.5)$$

First compute β :

$$\beta = \frac{\ln(100/80) - (0.10)(90/365)}{0.5} = \frac{\ln(1.25) - 0.02466}{0.5} \approx 0.3970.$$

Expected payoff in 90 days:

$$E(\max\{G_t - K, 0\}) = G_0 e^{t(\mu + \sigma^2/2)} P\left(Z > \frac{\beta - \sigma t}{\sqrt{t}}\right) - K P\left(Z > \frac{\beta}{\sqrt{t}}\right) \approx 1.788.$$

Since the option costs \$10 today, the **expected profit** is: $1.788 - 10 < 0$.

Hence, under this model, you should expect to **lose money** by buying this option at a price of \$10.

In this example, the initial option price \$10 was chosen arbitrarily.

A central problem in mathematical finance is how to determine a *fair price* for such options.

This leads to the Black–Scholes option pricing model, which is introduced in the next section.